Problem Set 3

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1 Q1: Regression of *voteshare* on *difflog*

We estimate:

$$\widehat{voteshare} = \beta_0 + \beta_1\, difflog.$$

Fitted equation:

$$\widehat{voteshare} = 0.5790 + 0.0417 \cdot difflog.$$

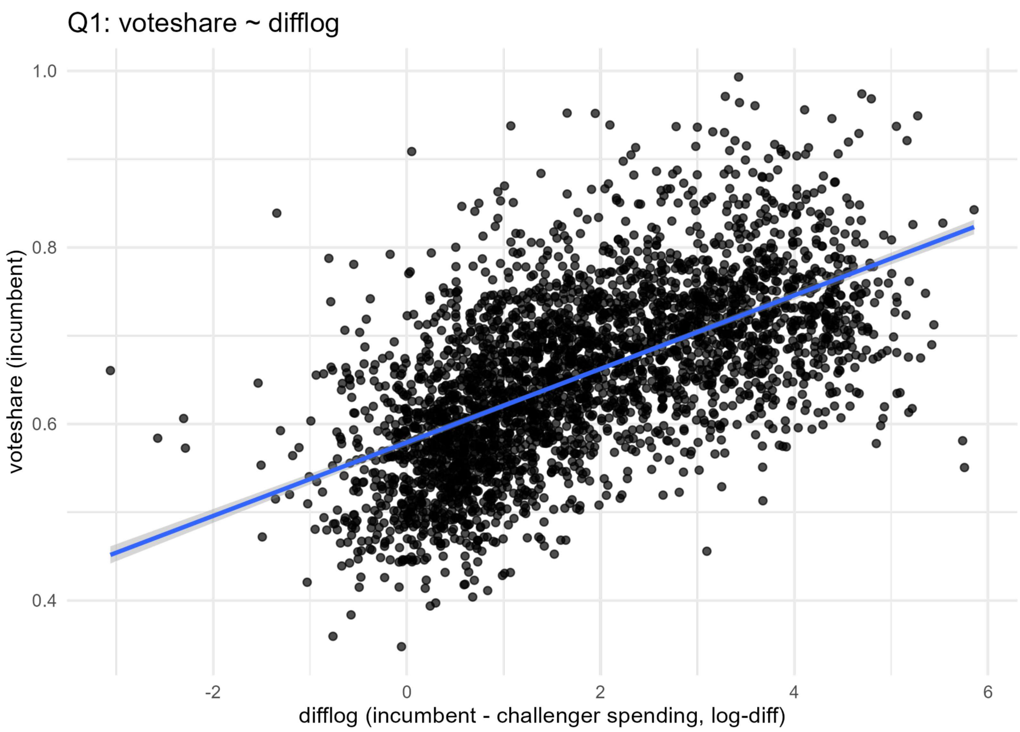

Figure 1: Q1: *voteshare* vs *difflog* with OLS fit.

2 Q2: Regression of *presvote* on *difflog*

We estimate:

$$\widehat{presvote} = \alpha_0 + \alpha_1\, difflog.$$

Fitted equation:

$$\widehat{presvote} = 0.5076 + 0.0238 \cdot difflog.$$

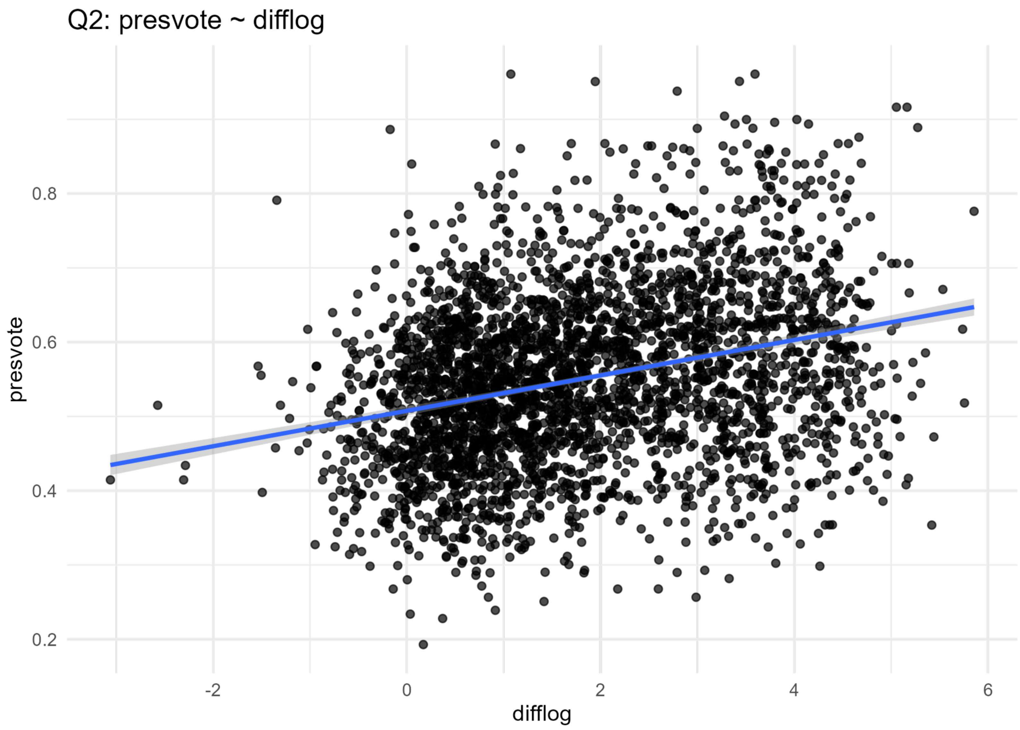

Figure 2: Q2: *presvote* vs *difflog* with OLS fit.

3 Q3: Regression of *voteshare* on *presvote*

We estimate:

$$\widehat{voteshare} = \gamma_0 + \gamma_1\, presvote.$$

Fitted equation:

$$\widehat{voteshare} = 0.4413 + 0.3880 \cdot presvote.$$

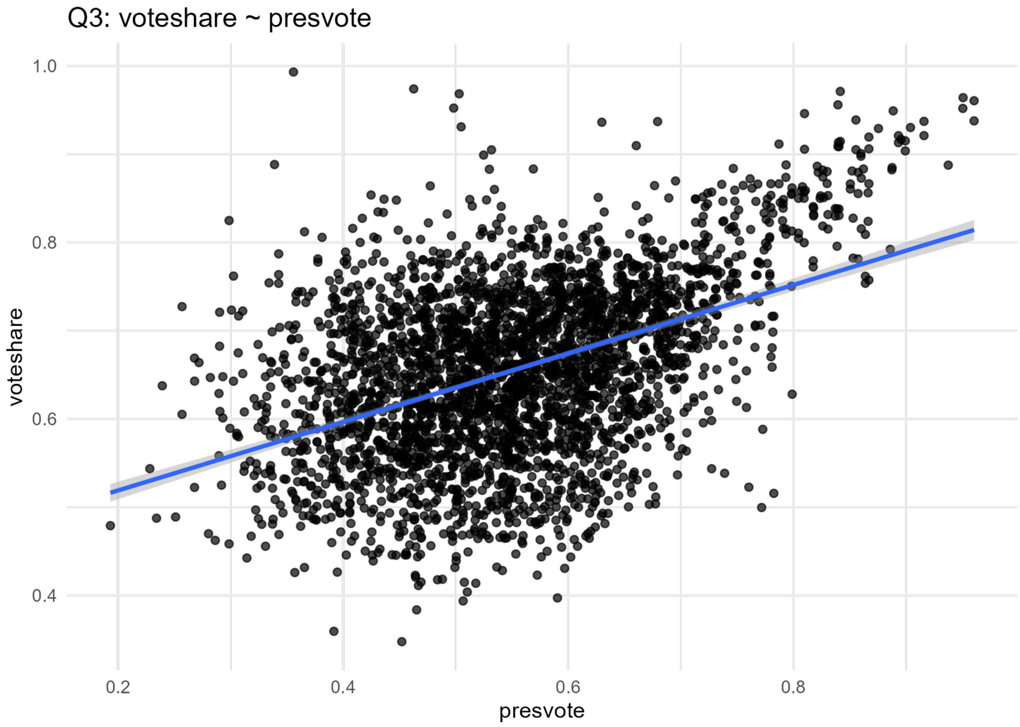

Figure 3: Q3: *voteshare* vs *presvote* with OLS fit.

4 Q4: Residual-on-Residual (Add-Variable) Regression

Let res_q1 be residuals from regressing *voteshare* on *difflog*, and res_q2 be residuals from regressing *presvote* on *difflog*. We estimate:

$$res\_q1 = \delta_0 + \delta_1 \, res\_q2 + \varepsilon.$$

Fitted equation:

$$\widehat{res\_q1} = -0.000000 + 0.256877 \cdot res\_q2.$$

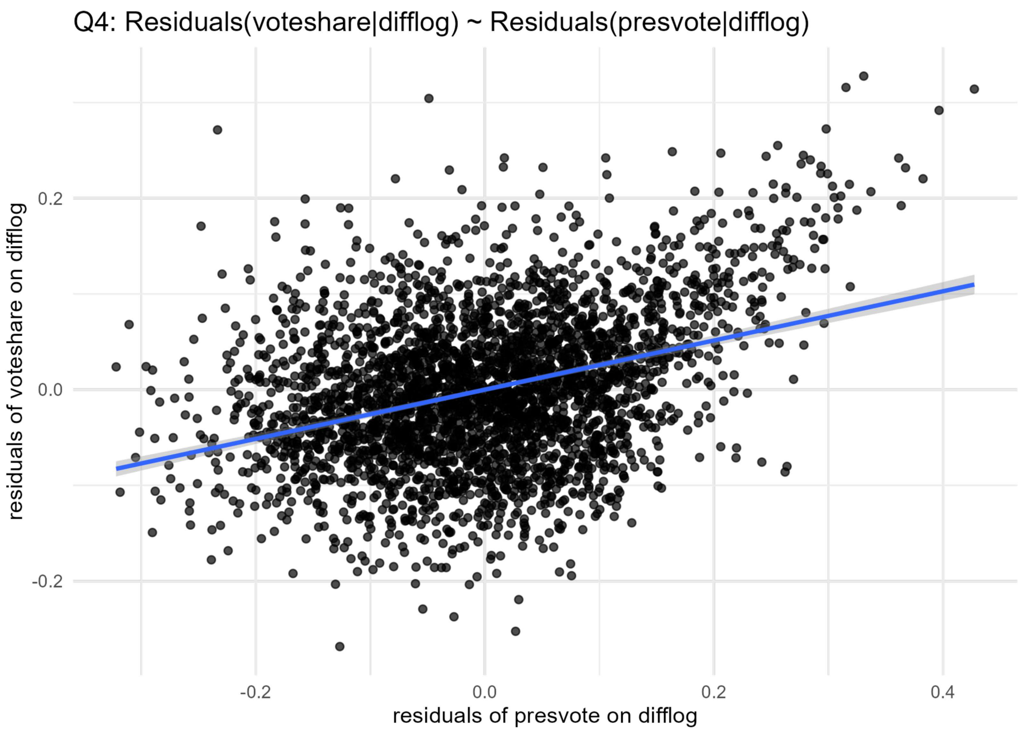

Figure 4: Q4: Added-variable plot – residuals of *voteshare* on *difflog* vs residuals of *presvote* on *difflog*, with fitted OLS line.

5 Q5: Multiple Regression and FWL Check

We estimate:

$$\widehat{voteshare} = \theta_0 + \theta_1\, difflog + \theta_2\, presvote.$$

Fitted equation:

$$\widehat{voteshare} = 0.4486 + 0.0355 \cdot difflog + 0.2569 \cdot presvote.$$

FWL Equality (Numerical Verification)

The coefficient on *presvote* in Q5 must equal the slope in Q4 for res_q2.

Table 1: FWL comparison: Q4 residual-on-residual vs Q5 multiple regression (coefficient on *presvote*).

Model	Estimate	Std. Error	t-statistic	p-value
Q4 residual-on-residual	0.2569	0.0118	21.8398	0.0000
Q5 multiple OLS	0.2569	0.0118	21.8363	0.0000

Interpretation. After controlling for *difflog*, *presvote* remains a strong predictor of *voteshare*. The identical entries across Q4 and Q5 validate the Frisch–Waugh–Lovell theorem.